

# INFORMATION TECHNOLOGY MANAGEMENT

## The Social Network For Fashion

### Wardrobe

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# Project Charter

## 1. Project Description

Wardrobe is a mobile-first app for discovering fashion products from different brands in one place. It uses a TikTok-style experience, where users browse a personalized feed of swipeable fashion items (clothes, shoes, bags, accessories). The goal is to make fashion discovery faster and more engaging than browsing multiple webshops manually in the same place.

The long-term vision is to build a social network for fashion, where users can interact, share products, and follow trends. The platform is intended to be monetized through affiliate marketing.

For the MVP, the focus is on discovery, personalization, saving favorites, and sharing products. The core flow is: swipe feed, product details, saving favorites, and opening the original product page on the brand website, with a basic way to import and update product data.

In the initial development phase, the app will use static mock data stored locally in JSON files. No live data integration or external data ingestion will be implemented at this stage.

## 2. Project Objectives

### A) User functionalities

- Registration/login (email + password is enough for MVP)
- Onboarding preferences (categories, style, price range, optional favorite brands)
- Feed (product cards, like, save, share products)
- Swipe feed (up=next, down =back, left=chat/profile)
- Product card details: image, brand, item name, price, category
- Product detail screen/page with more information and external link
- Wishlist / Saved items screen
- Search bar (category, brand, price range)

### B) Admin / back-office (basic)

- Product import/update from approved data source (CSV/API/feed/permitted method)
- Basic admin page/dashboard for product count, categories, brands, import status

- Manual refresh/import trigger (if possible)
- Basic usage analytics (users, swipes, likes, top categories/brands)

Note: This Admin dashboard is optional since for the MVP we will use mock data. But it will work as a prototype for clients to have a better understanding of how Wardrobe works.

### **C) Non-functional requirements**

- Mobile-first responsive UI
- Smooth and fast feed loading
- Secure authentication and password storage
- Clean code and documentation for handover
- Basic error handling and logging

## **3. Delivery Plan And Deadlines (Phased Approach Preferred)**

### ***Phase 1 - Planning and design (1–14 March 2026)***

Deliverables:

- Technical proposal (stack, architecture, data ingestion method, compliance note)
- Wireframes / clickable mockup
- Database schema draft
- Milestone plan

### ***Phase 2 - Core MVP development (15 March – 11 April 2026)***

Deliverables:

- Authentication
- Onboarding/preferences
- Swipe feed UI
- Product detail page
- Wishlist
- Backend API + database integration

### ***Phase 3 - Data pipeline + personalization + admin tools (12–30 April 2026)***

Deliverables:

- Product import/update pipeline
- Basic recommendation/personalization logic (rule-based is acceptable)
- Minimal admin dashboard
- Basic analytics counters

### ***Phase 4 - Testing, fixes, deployment, handover (1 – 31 May 2026)***

Deliverables:

- Bug fixes and QA testing
- Staging deployment / demo-ready version
- Source code handover
- Setup and deployment documentation

## **4. Available Resources (Cash And In-Kind)**

**Total budget available:** 10,000 EUR

**Allocation:**

- Development (frontend + backend): 5,000 EUR
- Hosting/tools/domains/testing: 1000 EUR
- Contingency: 1500 EUR

**In-kind/other assets:**

- Team effort: ~540 total hours (3 people × 180 hours).
- Software licenses are provided by the university.

If the budget is limited, priority should be given to core features only.

## **5. Key Personnels**

**Project team (students):**

- Aleeksandr Sokolov – Project management and coordination, deployment, production, integration.
- Viktor Taseski – Product vision, scraping/ingestion pipeline, data schema.
- Mario Krajoski – Frontend UI/UX, swipe interaction, client-side state.

## **6. Other Possible Partners**

The project may collaborate with:

- Professors and assistants
- Big brands (Zara, Bershka, Gymshark, Lululemon, etc.)
- Other UPR students can be involved to perform tests
- Tools, instruments and materials needed for the development



# Personas and Problem Scenarios

## Lea the Scroll-to-Style Explorer

### Screening Question:

When you discover fashion items, do you mostly use your phone, prefer quick visual browsing (scrolling/swiping), save items you like, and open the store page only when something really interests you?

### Persona Description:

Lea is a 21-year-old student who uses her phone as her main device for fashion discovery. She enjoys trends and visual inspiration, but she dislikes jumping between many web-shops and losing track of items she liked. She wants a fast, personalized experience that feels more like social media than traditional online shopping. She often browses for inspiration, saves favorites, and decides later whether to buy. This fits Wardrobe’s MVP well because it focuses on a swipe-based feed, personalization, favorites, sharing, and linking to the original store page.

|               |                                                                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Thinks</b> | <b>“Fashion discovery should be quick, fun, and tailored to my style.”</b>                                                                                                    |
| <b>Sees</b>   | Too many scattered web-shops, repeated products, irrelevant recommendations, and endless scrolling across different apps.                                                     |
| <b>Feels</b>  | Excited when she instantly finds something that matches her taste, but frustrated when shopping becomes slow, messy, or overwhelming.                                         |
| <b>Does</b>   | Browses in short mobile sessions, swipes quickly through products, saves favorites, shares items with friends, and opens the brand page when she is ready to consider buying. |

### Problem Scenario & Value Proposition

| <b>Problem Scenarios</b>             | <b>Current Alternatives</b> | <b>Your Value Proposition</b>   |
|--------------------------------------|-----------------------------|---------------------------------|
| Lea wants to discover fashion across | Opening multiple web-       | Wardrobe offers a mobile-first, |

|                                                                                                        |                                                                                                 |                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| multiple brands but does not want to open many different web-shops.                                    | shops, browsing Instagram, searching manually on Google, following influencers.                 | swipe-based feed that brings products from different brands into one place and improves through likes and skips.                   |
| Lea feels bored and wants to buy something but she does not know exactly what item she is looking for, | Random scrolling on Instagram or TikTok, visiting random web-shops, browsing without direction. | Wardrobe shows one product at a time in a personalized yet slightly unpredictable feed, making discovery engaging and exploratory. |
| Lea often saves items but loses track of them across different apps and stores                         | Taking screenshots, saving links in notes, bookmarking multiple sites.                          | Wardrobe allows her to save favorites in one organized place and easily return to them when she is ready to buy.                   |
| Lea values social influence and likes seeing what her friends are interested in.                       | Sending screenshots in group chats, asking friends directly, sharing Instagram posts.           | Wardrobe enables sharing, liking, and seeing what friends save, creating a social and more trustworthy discovery experience.       |

## Mark, The occasional shopper

### Screening Question:

Do you only buy clothes or accessories when you need them rather than regularly browsing fashion content?

### Persona Description:

Mark is a 35-year-old office worker. He dresses neatly but practically—jeans, polo shirts, casual sneakers, or dress shoes depending on the occasion. He doesn't actively follow fashion trends but wants to look presentable and stylish when needed. Mark often forgets which stores have what he wants, and he finds it frustrating to scroll through endless online catalogs. He values simplicity,

efficiency, and curated recommendations. He's more likely to shop when an event, work meeting, or special occasion requires it. He usually shops on weekends and spends as little time as possible online searching for the right item.

|               |                                                                                                                                                                                                                            |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Thinks</b> | <b>“Find the things you need quickly and efficiently, without spending unnecessary time.”</b>                                                                                                                              |
| <b>Sees</b>   | Complex websites with overwhelming product lists and ads for items he doesn't care about, as well as friends sharing trendy fashion that doesn't feel relevant to him.                                                     |
| <b>Feels</b>  | Frustrated by the time-consuming nature of online shopping but relieved and satisfied when products are easy to find.                                                                                                      |
| <b>Does</b>   | Shops online only one or two times per month, primarily for work or special occasions, using search filters and curated lists to quickly narrow options, and saves or purchases only the items that clearly fit his needs. |

### Problem Scenario & Value Proposition

| Problem Scenarios                                                                                                                                                                                                                                                                                                             | Current Alternatives                                                                                                                                                                                                                                                      | Your Value Proposition                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mark has an upcoming work event and needs a formal shirt. Since he shops only occasionally, he finds it frustrating to search through multiple online stores filled with hundreds of similar items. The process takes more time than he is willing to spend, and he often feels unsure whether he has chosen the best option. | Mark typically searches for shirts using Google or visits a few familiar online fashion stores. He manually browses categories, applies filters, and compares prices and styles across different websites. This approach requires opening many tabs, navigating different | Wardrobe provides Mark with a personalized and curated feed of formal shirts based on his preferences. Instead of manually browsing multiple stores, he can quickly swipe through relevant options, view essential product details, and open the original store page when ready to purchase, significantly reducing the time and effort required. |

|                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                             | interfaces, and repeatedly entering search criteria.                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                 |
| Mark wants to buy casual clothes for everyday or weekend use but does not actively follow fashion trends or social media influencers. He often delays shopping because browsing through large product catalogs feels unnecessary and overwhelming, even when he clearly needs new clothes.. | In this situation, Mark usually returns to the same few online stores he already knows or buys items without much comparison. Sometimes he waits until a physical store visit becomes necessary. These alternatives limit his exposure to better options and still require time and effort, which discourages him from shopping efficiently. | Wardrobe offers Mark a simplified discovery experience by presenting a small, relevant selection of casual clothing tailored to his style and budget. The swipe-based interface allows him to find suitable items quickly without engaging with trends, making occasional shopping faster and more comfortable. |

### Inna, the social fashion connector

#### Screening Question:

“Do you often share fashion items with friends, ask for opinions before buying, or send links/screenshots of outfits in chats?”

#### Persona description:

Inna is 20 years old and studies Communications. She is very active socially and enjoys discussing outfits and trends. Fashion for her is not only about buying clothes, but also about interaction and experience sharing.

She frequently sends screenshots of items to group chats, asks for opinions before purchasing, and shares links. She enjoys discovering items, but what excites her most is reacting to them together with others.

She is happy with the social side of fashion discovery. She wants to see what others like, what is trending among her circle, and what products are being shared.

|               |                                                                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Thinks</b> | <b>“Fashion is more fun when I can share it and see what others think.”</b>                                                                        |
| <b>Sees</b>   | Friends posting outfits, sending product links, reacting to styles, and influencing each other’s choices.                                          |
| <b>Feels</b>  | Excited when friends approve her picks, curious about what others are liking, and confused when she shops alone without interaction.               |
| <b>Does</b>   | Shares product links in chats, screenshots items, asks for feedback before buying, reacts to friends’ fashion posts, saves items to discuss later. |

### Problem Scenario & Value Proposition

| Problem Scenarios                                                                                                                                        | Current Alternatives                                                                                                  | Your Value Proposition                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Inna finds a fashion item she likes but wants feedback from friends before buying. Sharing links between different websites feels messy and unorganized. | Sending screenshots in group chats. Copy-pasting product links from multiple webshops. Posting Instagram story polls. | Our app allows users to share products directly inside the app. Friends can react, comment, or like items in one place.                       |
| Inna is curious about what fashion items her friends are saving or liking. She wants to see trends within her society.                                   | Asking friends directly in chats. Scrolling through social media posts. Meeting in person to discuss shopping ideas.  | Wardrobe introduces a social feed where users can see what friends like, save, or share.                                                      |
| Inna enjoys discovering new brands but prefers discovering them through people she trusts rather than random ads.                                        | Following influencers on Instagram. Browsing hashtags. Clicking targeted ads.                                         | The app combines personalization with social signals (likes, shares, saves from friends). This makes discovery more relevant and trustworthy. |

# User stories and test cases

## Lea the Scroll-to-Style Explorer

| User story                                                                                                                                                                      | Test cases                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lea wants to quickly swipe through a personalized fashion feed on my phone, so that discovering items feels fast, fun, and tailored to my style without opening many web-shops. | <b>Personalization reacts to behavior</b><br>Given Lea likes several “streetwear” items, when she continues swiping, then the feed should show more similar items (style/category/brands) within the next cards.<br><br><b>No duplicates</b><br>Given Lea has already seen Product X, when she swipes for a while, then Product X should not reappear (unless there’s an explicit “Seen again” feature). |
| As Lea, I want to save items I like into one organized favorites list, so that I don’t lose track of them across different apps and can come back later when I’m ready to buy.  | <b>Save from feed</b><br>When Lea taps “Save” on a product, then the item should appear in Favorites immediately and show a “Saved” state on the card.<br><br><b>Favorites persistence</b><br>Given Lea saved items yesterday, when she reopens the app today, then her favorites should still be there (same count and items).                                                                          |
| As Lea, I want to share items with friends and see what my friends are saving/liking, so that discovery feels social and I trust choices more.                                  | <b>Share an item</b><br>When Lea taps “Share” on a product, then the share sheet should appear and include a link that opens the product in Wardrobe (or store if Wardrobe isn’t installed).                                                                                                                                                                                                             |

|  |                                                                                                                                                                                         |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Friend activity visibility</b><br>Given Lea follows a friend, when that friend saves an item, then Lea should see it in a “Friends activity” area (with friend name + item preview). |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Mark, The occasional shopper

| User story                                                                                                                                                                               | Test cases                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| As Mark, an occasional shopper with limited time, I want to quickly find a formal shirt suitable for a work event, so that I don't have to browse multiple online stores and waste time. | <ul style="list-style-type: none"> <li>• Make sure formal shirts can be discovered through the personalized feed.</li> <li>• Make sure users can filter products by category (e.g., shirts) and style (formal).</li> <li>• Make sure each product clearly displays brand, price, and category.</li> </ul>                                                     |
| As Mark, who does not follow fashion trends, I want to see a small and relevant selection of casual clothes, so that shopping feels simple and not overwhelming.                         | <ul style="list-style-type: none"> <li>• Make sure casual clothing can be filtered by category.</li> <li>• Make sure the feed shows a limited and curated set of products instead of large catalogs.</li> <li>• Make sure products can be filtered by price range.</li> <li>• Make sure casual items shown are relevant to the user's preferences.</li> </ul> |
| As Mark, I want to save items that meet my needs, so that I can quickly return to them later without searching again.                                                                    | <ul style="list-style-type: none"> <li>• Make sure saved items are easily accessible from a dedicated screen.</li> <li>• Make sure users can remove items from their saved list.</li> <li>• Make sure saved items remain available after logging</li> </ul>                                                                                                   |

|  |                                                                                                                                                             |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>out and back in.</p> <ul style="list-style-type: none"> <li>• Make sure saved items retain correct product information (image, price, brand).</li> </ul> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Inna, the social fashion connector

| User story                                                                                                                                  | Test cases                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| As Inna, I want to create a quick poll on a product so my friends can vote if they like it or not, so that I can decide what to buy faster. | <ul style="list-style-type: none"> <li>• When Inna taps “Create poll” on a product, then a poll with options (e.g.,  /  or “Buy / Skip”) should appear for her friends.</li> <li>• During the voting, the vote count should update in real time.</li> <li>• When the poll ends, Inna should see the final results.</li> </ul> |
| As Inna, I want to comment on a product my friend shared, so that we can discuss outfits directly around the item.                          | <ul style="list-style-type: none"> <li>• When Inna opens a shared product, there should be a comment section attached to that item.</li> <li>• When she posts a comment, it should appear immediately under the product.</li> </ul>                                                                                                                                                                                                                                                                 |
| As Inna, I want to see trending items among my friends, so that I can discover what people in my circle currently like.                     | <ul style="list-style-type: none"> <li>• When several friends interact with the same item, the product should appear in a special section (say: “My friends favorites”).</li> <li>• The section should display the number of friends interacting with the item.</li> </ul>                                                                                                                                                                                                                          |



# Hypothesis testing

## People that we interviewed

| Name                | Age | Job / Occupation                            |
|---------------------|-----|---------------------------------------------|
| Stefan Petrovski    | 23  | Student in Skopje, N. Macedonia             |
| Antonio Dimitrieski | 34  | Mechanical engineer in Skopje, N. Macedonia |
| Anita Osmani        | 22  | UP student from N. Macedonia                |
| Lea Stankovic       | 22  | UP student from Serbia                      |
| Sergej Savinkov     | 21  | UP student from Russia                      |

## Questions that were asked during the interview:

- How often do you shop for clothes online?
- What platforms or websites do you usually use to discover fashion items?
- What problems do you experience when shopping for clothes online?
- Do you find it difficult to search across multiple fashion stores?
- Would you use an app that shows clothing items from many brands in one swipe-based feed?
- How important are personalized recommendations when browsing clothing?
- Would you share fashion items with friends before buying them?

## Answers:

### From Person 1 – Stefan Petrovski

1. **Q:** How often do you shop for clothes online?

**A:** I usually shop online 3-5 times a month. That is because I want to see what is popular and sometimes few things can catch my eye.

2. **Q:** What platforms or websites do you usually use to discover fashion items?

**A:** Typically, I search on Google, or I check Temu since there are a lot of cheap things. From time to time, I visit a few known sites, for example Pull & Bear or H&M etc.

3. **Q:** What problems do you experience when shopping for clothes online?

**A:** For example, I see a product that I like, but I forget in which category it was or which filters I applied in the menu. Also, the designs of the web stores can be really bad and therefore user experience is not satisfying.

4. **Q:** Do you find it difficult to search across multiple fashion stores?

**A:** Usually the answer is yes, especially when I don't know the exact product or product type that I want to find. Comparing similar items from 2 or more different stores is also time-consuming and sometimes frustrating/boring.

5. **Q:** Would you use an app that shows clothing items from many brands in one swipe-based feed?

**A:** I think it is much more user friendly and has a better approach, so yes, I would definitely give it a try.

6. **Q:** How important are personalized recommendations when browsing clothing?

**A:** Very important! As each person has his own style, that is the case with me too. I would like to see only items that match my everyday style and not waste time unnecessarily.

7. **Q:** Would you share fashion items with friends before buying them?

**A:** Sometimes yes. It happened a few times that the links did not actually work when I shared them.

## **From Person 2 – Antonio Dimitrieski**

1. **Q:** How often do you shop for clothes online?

**A:** I usually shop online every 3-4 months. That is because I have limited time in the day and I do not usually think about scrolling through the web stores.

2. **Q:** What platforms or websites do you usually use to discover fashion items?

**A:** Through social media mostly. For example, posts on Instagram or Facebook, or if I see information about sale in a particular clothing company, I open it after.

3. **Q:** What problems do you experience when shopping for clothes online?

**A:** Biggest problem is that is time consuming and not adaptable to the user's preferences or needs. If a person wears classic clothes or suits, it is not quite effective for him to see sport clothes.

4. **Q:** Do you find it difficult to search across multiple fashion stores?

**A:** Yes, that is overwhelming and difficult. It often happens that too much time and energy is spent on just searching.

5. **Q:** Would you use an app that shows clothing items from many brands in one swipe-based feed?

**A:** Of course, if it is well-designed, well-structured and fast regarding the whole process (from searching to purchasing).

6. **Q:** How important are personalized recommendations when browsing clothing?

**A:** On a scale from 1 to 10 I would say 10. Everyone wants to find items quickly and be happy about their choices. Additionally, it saves you a lot of time.

7. **Q:** Would you share fashion items with friends before buying them?

**A:** Usually, I share fashion items with my wife, so she decides if they are good. Sometimes I do not need feedback and just continue with the process, but I think it can be useful to share or save items that you like.

### **From Person 3 – Anita Osmani**

1. **Q:** How often do you shop for clothes online?

**A:** One or two times in six months.

2. **Q:** What platforms or websites do you usually use to discover fashion items?

**A:** Shien, Temu, Zalando, AboutYou, Notino, Bershka and Zara.

3. **Q:** What problems do you experience when shopping for clothes online?

**A:** I don't know if the size will fit, is the quality as good as in the pictures, are the colors the same in person and how long does shipping take.

4. **Q:** Do you find it difficult to search across multiple fashion stores?

**A:** Yes, switching through multiple websites is wasting my time and I am losing interest quickly and abandoning shopping.

5. **Q:** Would you use an app that shows clothing items from many brands in one swipe-based feed?

**A:** Yes, I would give it a try.

6. **Q:** How important are personalized recommendations when browsing clothing?

**A:** Yes, I love it when I receive recommendations. Since I find it easier to build an outfit that goes with previously bought clothes.

7. **Q:** Would you share fashion items with friends before buying them?

**A:** Absolutely, I would like to see what my friends or family think of the product I want to buy. I would not buy clothing that my friends do not like.

#### **From Person 4 – Lea Stankovic**

1. **Q:** How often do you shop for clothes online?

**A:** Not that often maybe 3-4times a year.

2. **Q:** What platforms or websites do you usually use to discover fashion items?

**A:** Pinterest, tiktoks and shein.

3. **Q:** What problems do you experience when shopping for clothes online?

**A:** The biggest one is with sizes, because I get the wrong one every time.

4. **Q:** Do you find it difficult to search across multiple fashion stores?

**A:** No, I do not.

5. **Q:** Would you use an app that shows clothing items from many brands in one swipe-based feed?

**A:** Yes, sounds interesting.

6. **Q:** How important are personalized recommendations when browsing clothing?

**A:** I think that they are important, on a scale 1-10 maybe a 7.

7. **Q:** Would you share fashion items with friends before buying them?

**A:** Yes, I often share items to friends before buying them

#### **From Person 5 – Sergej Savinkov**

1. **Q:** How often do you shop for clothes online?

**A:** I shop for clothes online a few times a year, usually when I need something specific like shoes, a jacket, or seasonal clothes.

2. **Q:** What platforms or websites do you usually use to discover fashion items?

**A:** I usually discover clothes on platforms like Instagram or TikTok, and sometimes directly on websites of popular brands or online stores.

3. **Q:** What problems do you experience when shopping for clothes online?

**A:** One common problem is that browsing through many products takes a lot of time. It can also be difficult to determine if the size will fit correctly.

4. **Q:** Do you find it difficult to search across multiple fashion stores?

**A:** Yes, it can be inconvenient because I have to open many different websites and search for the same item again and again

5. **Q:** Would you use an app that shows clothing items from many brands in one swipe-based feed?

**A:** Yes, I would probably use it because it would make browsing clothes faster.

6. **Q:** How important are personalized recommendations when browsing clothing?

**A:** They are quite important because they help find clothes that match my style and save time when searching.

7. **Q:** Would you share fashion items with friends before buying them?

**A:** Yes, sometimes I send links or screenshots to friends to ask for their opinion before buying something.

### **Insights:**

- Social media is the main discovery channel (80%).
- Most participants find it difficult and inconvenient to compare products across different stores (80%).
- The biggest issue is time spent browsing and comparing items, followed by size accuracy.
- **All** respondents showed interest in the proposed concept
- Personalization is highly valued because it saves time and matches personal style (~8.5/10).
- Sharing is common, especially with friends, partners, or family for feedback.

## Conclusions:

The interviews confirm our main hypothesis that users find it time-consuming and frustrating to browse multiple fashion websites when looking for clothing items. Participants expressed interest in the idea of a single app that aggregates products from different brands, which supports the core concept of the Wardrobe platform.

Another important finding is that personalized recommendations are very important to users. The interviewees said they prefer seeing items that match their style instead of browsing many irrelevant products.

The interviews also showed that simple navigation and fast browsing are essential, since users do not want to spend much time searching for clothes.

Overall, the results support our hypotheses about the Wardrobe concept. No major changes are required, but the app should focus strongly on personalization, simplicity, and quick product discovery.

**Prototype link [here](#).**

or



# Usability Testing

## Assessment plan

### **Objective:**

To evaluate whether users can easily use the Wardrobe concept to discover, filter, select and save clothing items efficiently.

### **Participants:**

- 1) Stefan Petrovski
- 2) Antonio Dimitrievski
- 3) Sergej Savinkov

### **Test method:**

Use low-fidelity prototype for the testing, providing few of the main functionalities of the application. The users will be evaluated by performing short tasks within the application. Time from start till completion will be monitored for each task that is performed by a specific user.

### **Test tasks:**

- Find a formal shirt using the app.
- Save an item to favorites and find it in closet.
- Open a product and go to the original store.



### **General plan for the execution of the testing**

| # | Activity                         | Duration (min) | Notes                                                                                                                                                                                              |
|---|----------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Intro & brief explanation        | 5              | Explain the main idea about the application, objectives, and goals regarding the testing. Ask participants to sign an online document which allows usage of their data collected from the testing. |
| 2 | Discovery questions for feedback | 5              | Ask for first impressions, thoughts and use the interview guide to improve the user stories or potentially add other additional properties to the previously done work.                            |
| 3 | Test tasks                       | 15             | Introduce test scenarios and ask participants to complete the previously defined tasks.                                                                                                            |
| 4 | Post-Test talk and debrief       | 5              | Ask some follow-up questions, make sure to get as much feedback as possible.                                                                                                                       |

#### **Success criteria:**

- Users complete tasks without confusion.
- Tasks are completed quickly (under 1 minute each).
- Users easily navigate through the menus of the application.
- Users can recognize and understand product information (price, brand, image)
- Users can easily save items to favorites.
- Users can quickly find the original store of a product.

#### **Results presentation:**

### *User 1 – Stefan Petrovski*

- Task 1 – 53 seconds – completed successfully, bit of confusion appeared at the beginning
- Task 2 – 20 seconds – no issues, completed easily
- Task 3 – 44 seconds – solved with no bigger issues

### *User 2 – Antonio Dimitrievski*

- Task 1 – 20 seconds – completed quickly, was trying to find a search bar and where to find products.
- Task 2 – 9 seconds – no issues, completed easily
- Task 3 – 21 seconds – tried to open it with share icon, eventually found the right button.

### *User 3 – Sergej Savinkov*

- Task 1 – 35 seconds – completed successfully, took some time to understand navigation but no major confusion
- Task 2 – 14 seconds – completed smoothly with minor hesitation
- Task 3 – 32 seconds – slight confusion between screens, but completed without assistance

## **Findings:**

Most users were able to complete all tasks successfully and without major confusion, which indicates that the application is intuitive. The easiest task was saving items to favorites, while the most difficult task was finding and using filters. Users spent slightly more time on the first task, which suggests that initial navigation can be

improved or indicated with animations at the first start of the application. Overall, users found the concept simple and efficient, but improvements are needed in filter visibility and navigation clarity.

# Scrum Basics and Jira Setup

## Introduction

We are using SCRUM as our agile framework. The core principle is to divide the project into small, manageable development cycles called sprints, rather than attempting to plan and implement the entire solution at once. This approach allows the team to maintain a clear objective within a fixed timeframe, minimize scope changes during the sprint, and conclude each cycle with a review and evaluation of the completed work.

Below is an example of how our sprints are structured in Jira, along with a template intended for future sprint planning and task organization.

### Sprint 1 — Project Foundation

| Task                                      | Priority | Story Points | Status |
|-------------------------------------------|----------|--------------|--------|
| Project repository setup & CI/CD pipeline | High     | 3            | Done   |
| Database schema design                    | High     | 5            | Done   |
| User authentication (register/login)      | High     | 8            | Done   |
| Basic UI layout & navigation              | Medium   | 5            | Done   |
| API architecture setup                    | High     | 3            | Done   |

### Sprint 2 — Core Features

| Task                          | Priority | Story Points | Status |
|-------------------------------|----------|--------------|--------|
| User profile management       | High     | 5            | Done   |
| Main dashboard view           | High     | 8            | Done   |
| Data input forms & validation | High     | 5            | Done   |
| REST API endpoints (CRUD)     | High     | 8            | Done   |
| Unit tests for core modules   | Medium   | 3            | Done   |

### Sprint 3 — Extended Functionality

| Task                             | Priority | Story Points | Status      |
|----------------------------------|----------|--------------|-------------|
| Search & filtering functionality | Medium   | 5            | In Progress |
| Notifications system             | Medium   | 8            | In Progress |
| Role-based access control        | High     | 8            | Review      |
| Reporting module (basic)         | Medium   | 5            | To Do       |
| Third-party API integration      | Low      | 13           | To Do       |

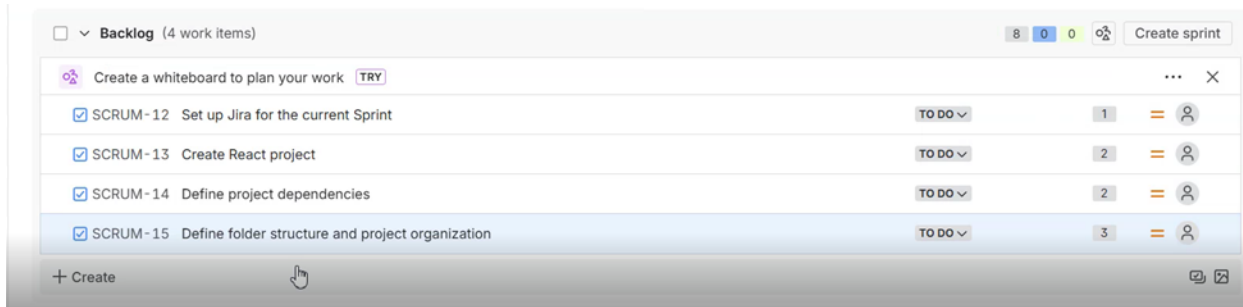
### Sprint 4 — Polish & Deployment

| Task                               | Priority | Story Points | Status |
|------------------------------------|----------|--------------|--------|
| End-to-end testing                 | High     | 8            | To Do  |
| Performance optimization           | Medium   | 5            | To Do  |
| UI/UX refinements                  | Medium   | 3            | To Do  |
| Documentation & user guide         | Low      | 3            | To Do  |
| Production deployment & monitoring | High     | 5            | To Do  |

The product backlog contains all planned user stories and tasks for the application.

Each task includes:

- Title
- Description
- Priority
- Story point estimation
- Assignee
- Status



## Estimation method

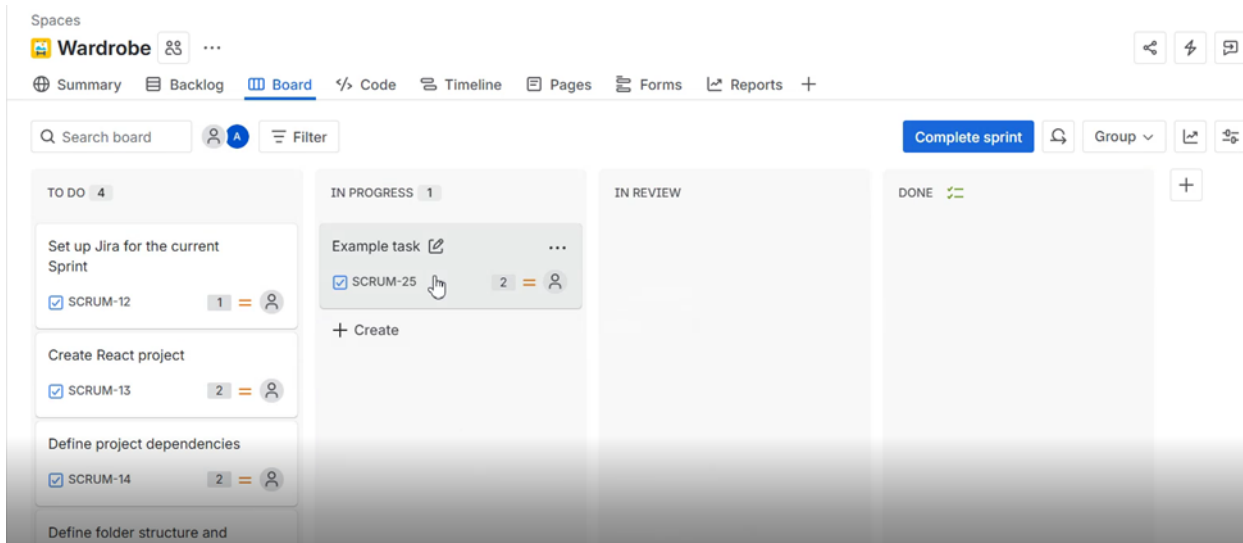
We used the Fibonacci sequence for task complexity estimation:

1, 2, 3, 5, 8, 13

This helps estimate development complexity and sprint workload.

## Scrum Board Workflow

The Scrum board was used to track task progress



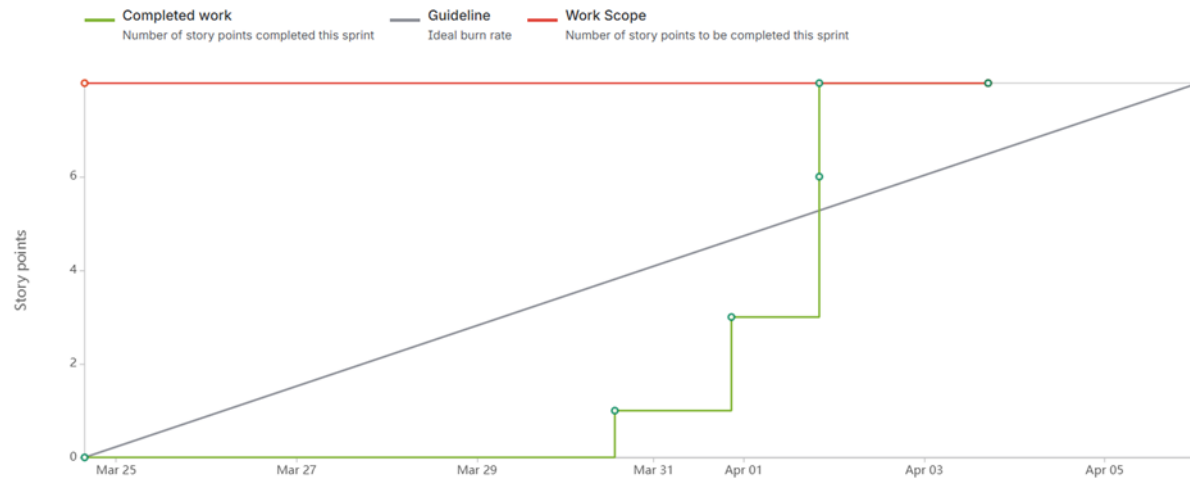
## Workflow Stages

1. **To Do** — tasks waiting to start
2. **In Progress** — tasks currently being developed
3. **Review** — tasks under testing or review
4. **Done** — completed tasks

This workflow improved transparency and helped track sprint progress effectively.

## Burndown Chart Analysis

**Sprint goal** - Migrate the hardcoded Wardrobe prototype into a React application with a clear folder structure, reusable components, separated styles, and base state management. Add some high-end features and mechanics.



Burndown charts were used to monitor sprint progress and compare completed work against the ideal guideline.

### Benefits of Burndown Charts

- Visual progress tracking
- Early detection of delays
- Sprint performance evaluation
- Better sprint planning for future iterations

The charts helped evaluate workload distribution and sprint efficiency.

# Project Stakeholders

## Online stores / product providers (Zara, Zalando, H&M)

**Status:** Positive

**Expectations / interests:**

These stakeholders are interested in getting more visibility, traffic, and possible sales through the app. They expect their product information and links to be shown correctly.

**Possible conflicts or benefits:**

They benefit if the app redirects users to their original store pages.

They may require promotion of priority, ads, or commissions, which could negatively affect user experience or app neutrality.

**Team response:**

The team should make sure product details are accurate; links are valid and not outdated. Also, accept all the related tradeoffs.

## Advertising partners

**Status:** Positive

**Expectations / interests:**

They want ad placements and user engagement to generate revenue.

**Possible conflicts or benefits:**

Benefit: Revenue source for the app.

Conflict: Too many ads can ruin user experience.



**Team response:**

Keep ads minimal and non-intrusive.

**Data protection / regulators**

**Examples:** European Union Regulators

**Status:** Neutral

**Expectations / interests:**

They expect user data to be handled securely and in compliance with privacy laws.

**Possible conflicts or benefits:**

Conflict: Strict regulations may limit data collection and tracking.

Benefit: Compliance increases user trust and credibility of the application.

**Team response:**

The team should minimize personal data usage, ensure transparency, and follow privacy regulations.

**End users (App users)**

**Status:** Positive

**Expectations / interests:**

Users want a fast, simple, and personalized way to discover clothing items without browsing multiple websites. They expect accurate product information and a smooth user experience.

**Possible conflicts or benefits:**

Benefit: Easier and faster product discovery.

Conflict: Too many ads or irrelevant recommendations may reduce user satisfaction.

**Team response:**

Focus on simplicity, personalization, and performance. Ensure minimal ads and relevant recommendations.

# Positive patterns

## 1. Be Honest

### Description

Transparency and honesty help create trust inside the team and with users.

### Application in Wardrobe

Wardrobe redirects users to external stores and uses recommendation systems. The team must communicate clearly about sponsored content, product availability, and technical limitations.

### Benefits

- Increased trust inside the team
- Better communication with users
- Stronger long-term reputation

Team Approach

### We would:

- Clearly label sponsored products
- Share project progress honestly within the team
- Encourage constructive feedback and transparency during development

## 2. Set Clear Goals

### Description

Teams work more effectively when expectations and objectives are clearly defined.

### Application in Wardrobe

Our project includes recommendation systems, user interface improvements, and external store integrations. Without clear goals, priorities may become confusing.

### Benefits

- Better team coordination
- Easier prioritization
- Improved project planning
- Reduced misunderstandings

## Team Approach

### **We would:**

- Define sprint goals clearly and early
- Break large tasks into measurable milestones
- Use project management tools to track progress
- Ensure every team member understands the goals clearly

## **3. Be a Teacher and a Mentor**

### **Description**

Strong engineering teams grow when experienced members help others learn, improve technical skills, and gain confidence.

### **Application in Wardrobe**

Our project combines frontend development, backend systems, APIs, and recommendation algorithms. Team members may have different levels of experience in these areas, so knowledge sharing is important.

### **Benefits**

- Faster skill development inside the team
- Better code quality and engineering practices
- Stronger collaboration and team culture

## **Team Approach**

We would:

- Use pair programming and collaborative code reviews
- Encourage more experienced members of the group to mentor junior members
- Share useful resources and technical knowledge regularly
- Create a supportive environment (e.g. chats) where asking questions is encouraged

# Negative patterns

## 1. Poor Communication and Lack of Transparency

### Description

When team members do not communicate openly, misunderstandings, hidden problems, and confusion can appear during development.

### Application in Wardrobe

Since Wardrobe includes frontend, backend, APIs, and recommendation systems, weak communication between team members could cause integration problems and delays.

### Risks

- Misunderstandings between developers
- Delayed task completion
- Increased number of bugs and mistakes
- Lower team motivation

### Team Approach

We would:

- Hold short weekly progress meetings
- Share problems early instead of hiding them
- Use tools like Trello or Jira for transparent task tracking
- Encourage open and respectful communication inside the team

## 2. Unclear Roles and Responsibilities

### Description

If team members do not know exactly what they are responsible for, tasks may be duplicated or ignored completely.

### Application in Wardrobe

Different parts of the project require different skills. Without clear responsibilities, important tasks such as API integration, UI design, or testing may be forgotten.

### Risks

- Confusion inside the team
- Uneven workload distribution
- Slower development process
- Reduced accountability

### Team Approach

We would:

- Assign clear responsibilities to each team member
- Define roles early in the project
- Review responsibilities regularly during development
- Make sure everyone understands their tasks and deadlines

## 3. Lack of Knowledge Sharing

### Description

When knowledge stays with only one person, the team becomes dependent on specific members and collaboration becomes weaker.

## **Application in Wardrobe**

If only one person understands the recommendation system or backend logic, the project may slow down when that person is unavailable.

## **Risks**

- Dependency on individual team members
- Slower problem solving
- Difficult onboarding for new members
- Lower overall team efficiency

## **Team Approach**

We would:

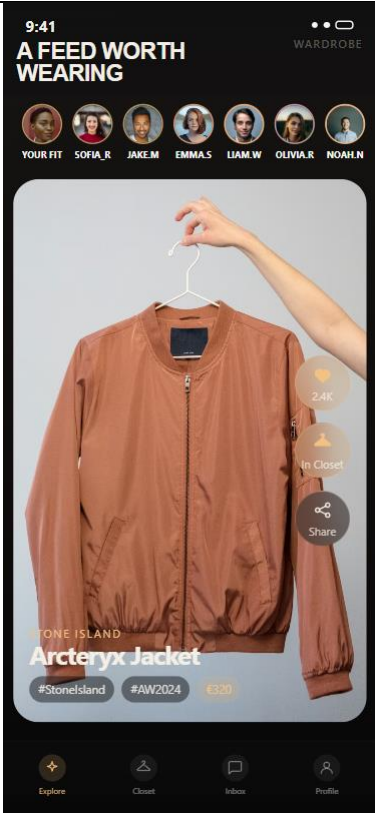
- Organize code reviews and pair programming
- Share technical documentation with the team
- Encourage mentoring and collaborative learning
- Regularly explain important technical decisions to all members



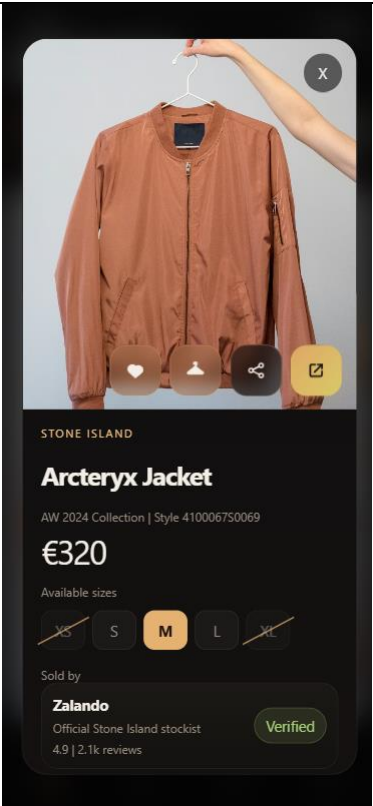
# Showcases of the final app

The final Wardrobe prototype demonstrates a modern mobile-first fashion discovery experience inspired by social media platforms. The application focuses on simplicity, speed, personalization, and social interaction while helping users discover clothing items from multiple brands in one place.

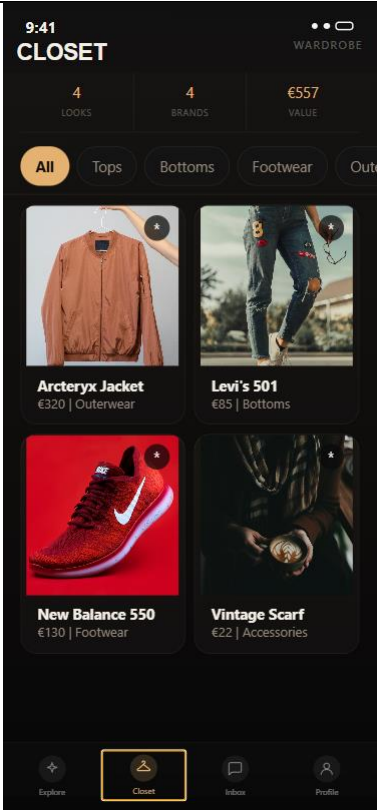
The interface follows a clean dark-themed design optimized for smartphones. Large product images, minimal distractions, and easy navigation help users focus on the content itself.

| Showcase                  | Image Placement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Showcase 1 — Explore Feed |  A screenshot of the 'Wardrobe' app's 'Explore Feed' interface. The screen is dark-themed. At the top, the time is 9:41 and the app name 'WARDROBE' is in the top right. Below the header, there's a section titled 'A FEED WORTH WEARING' with a row of seven circular profile icons labeled 'YOUR FIT', 'SOFIA_R', 'JAKE.M', 'EMMA.S', 'LIAM.W', 'OLIVIA.R', and 'NOAH.N'. The main content area features a large image of a brown bomber jacket hanging on a white hanger, held by a hand. To the right of the jacket image are three circular icons: a heart with '2.4K', an upward arrow with 'In Closet', and a share icon with 'Share'. Below the jacket image, the text 'STONE ISLAND' is in small letters, followed by 'Arcteryx Jacket' in a larger font. At the bottom of the post are two tags: '#StoneIsland' and '#AW2024', and a price tag '€320'. The bottom navigation bar has four icons labeled 'Explore', 'Closet', 'Inbox', and 'Profile'. |

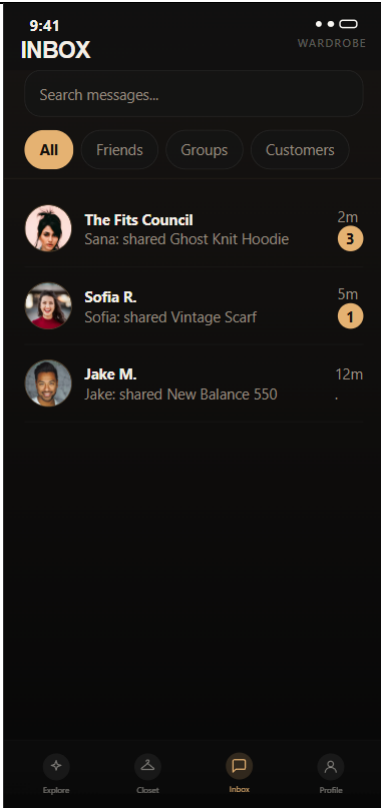
Showcase 2 — Product Details Screen



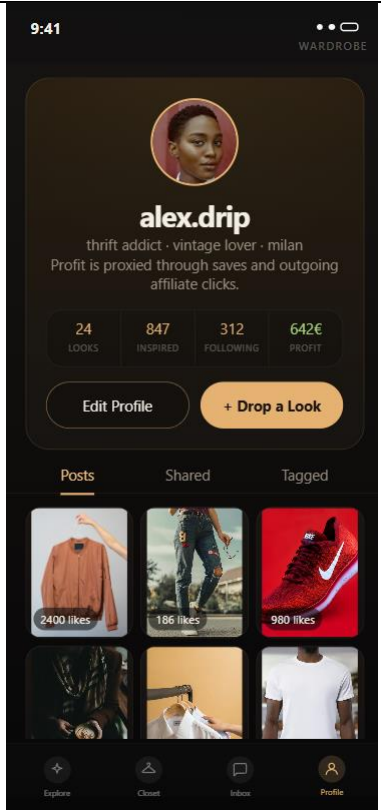
Showcase 3 — Closet / Saved Items



Showcase 4 — Social Features / Inbox



Showcase 5 — Profile



## Conclusion

The MIT course provided practical experience in software project management, agile development, teamwork organization, and user-centered design. Throughout the project, we applied concepts such as SCRUM methodology, Jira task management, stakeholder analysis, user research, usability testing, and project planning in a real development scenario.

The course helped us better understand how software projects are structured and managed from the initial idea to prototype development, while also improving collaboration, communication, and problem-solving skills within a development team.

As a final result, we successfully designed and developed the Wardrobe prototype, which demonstrates the core concept of a modern social fashion discovery platform with personalized recommendations, swipe-based navigation, and social interaction features.